

Test:- ML and Statistics Theoretical(19th July)

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Q5)Ans:- 10 key differences between descriptive statistics and inferential statistics:

1. Definition:

- **Descriptive Statistics:** Summarizes and describes the main features of a dataset.
- **Inferential Statistics:** Makes predictions or inferences about a population based on a sample of data.

2. Purpose:

- **Descriptive Statistics:** Provides simple summaries and visualizations of the data.
- **Inferential Statistics:** Helps in making generalizations and decisions based on the data.

3. Data Scope:

- **Descriptive Statistics:** Deals with data from a specific dataset or population.
- **Inferential Statistics:** Uses sample data to infer about the larger population.

4. Techniques:

- **Descriptive Statistics:** Includes measures like mean, median, mode, range, variance, standard deviation, and visualizations like histograms and pie charts.
- **Inferential Statistics:** Includes hypothesis testing, confidence intervals, regression analysis, and ANOVA.

5. Complexity:

- **Descriptive Statistics:** Generally simpler and easier to calculate.
- **Inferential Statistics:** More complex, often involving more advanced mathematical calculations.

6. Focus:

- **Descriptive Statistics:** Focuses on presenting raw data in a meaningful way.
- **Inferential Statistics:** Focuses on drawing conclusions and making predictions about a population.

7. Outcome:

- **Descriptive Statistics:** Produces exact values or visual representations of the data.
- **Inferential Statistics:** Produces probabilities and estimations about the population.

8. Assumptions:

- **Descriptive Statistics:** Does not make assumptions about the population from which the data is drawn.
- **Inferential Statistics:** Often makes assumptions about the population and relies on the sampling methods.

9. Sample Size:

- **Descriptive Statistics:** Can work with the entire population or a sample without needing to generalize.
- **Inferential Statistics:** Relies on sample data to make inferences about the larger population, making sample size crucial.

10. Examples:

- **Descriptive Statistics:** Calculating the average test score of students in a class, creating a bar chart of sales figures.
- **Inferential Statistics:** Estimating the average height of all students in a school based on a sample, testing if a new drug is more effective than a placebo.

Q6) Ans:- To handle missing values in a dataset, we can use several methods:

1. **Drop Missing Values:** Remove rows or columns with missing data using `dropna()`.
2. **Fill Missing Values:**
 - **Mean/Median/Mode Imputation:** Replace missing values with the mean, median, or mode of the column using `fillna()`.
 - **Forward/Backward Fill:** Use the previous or next value to fill the missing values with `ffill()` or `bfill()`.
3. **Imputation with Algorithms:** Use more sophisticated methods like `KNNImputer` or `IterativeImputer` from `sklearn` for filling missing values based on other data points.